

Tea Factory !' Tea Marketplace

API Integration Reference

TokenHarvest · East African eWR Trading Platform · v1.0

Tea Factory !' Tea Marketplace API Integration

Base path: /api (all endpoints below are relative to this prefix)

Authentication: Clerk session cookie (web) or Bearer token (mobile). Every endpoint that mutates state requires a valid Clerk session.

Content-Type: application/json for all requests and responses.

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Overview

The Tea Factory integration connects tea producers and their mandate-holding brokers to the Tea Marketplace auction and spot-market system. The flow is:

```
Tea Producer (owner)
  % % grants mandate to Broker (ENABLER)
    % % Broker creates Tea Lots linked to producer's eWR
      % % Broker compiles Lots into an Auction Session
        % % Buyers (OFF_TAKER) place bids in real time
          % % Worker closes lots !' SOLD or RESERVE_NOT_MET
            % % Admin confirms payment !' Delivery Order issued
```

Each tea lot is backed by a Tea Electronic Warehouse Receipt (eWR) of commodityType = "TEA". When a lot is sold, the eWR ownership is atomically transferred to the winning buyer and its state is set to SETTLED.

User Roles

Tier	Who	Capabilities
------	-----	--------------

ENABLER	Tea broker	Create lots, create & start auction sessions, take-out / accept-below-reserve
OFF_TAKER	Tea buyer	Place bids, view settlement details for lots they won
ADMIN	Platform operator	All broker actions + confirm payment settlement
(owner)	Tea producer	View and update their own lots; attach dispatch docs

Role is stored on the users table as tier. The system resolves the caller's role from their Clerk session via `getAuth(req)`.

Mandate System

Before a broker can catalogue a tea lot, the producer (eWR owner) must grant them an active broker mandate for commodity type "TEA".

A mandate is active when:

- revoked = false
- validFrom has passed (or is null)
- validTo has not yet passed (or is null)

If `commissionRateOverride` is set on the mandate, it overrides the `commissionRate` the broker specifies at lot-creation time.

Without an active mandate, POST /tea/lots returns 403.

Lot Lifecycle

```

DRAFT
% (PATCH updates metadata, triggers CATALOGUED)
%4
CATALOGUED
% (POST /dispatch with PRE_AUCTION_DISPATCH)
%4
DISPATCHED
% (session created, session started)
%4
LIVE % % % % auctionEndAt expires % % % % %° SOLD (reserve met)
% % % %° RESERVE_NOT_MET (reserve not met)
%
% % % % % % % %4% % % % % % % %
%4
WITHDRAWN (accept-below-reserve)
%
%4
SOLD

```

Lots can only be edited (PATCH) while in DRAFT or CATALOGUED status.

Session Lifecycle

```

SCHEDULED
% (POST /tea/auctions/:id/start)
%4
LIVE % % % % worker closes last lot % % %° CLOSED

```

A session holds an ordered catalogue of lot IDs. The worker advances through them automatically, one at a time, after each lot closes.

Endpoints — Lots

POST /tea/lots

Create a new tea lot catalogue entry. Requires an active mandate from the eWR owner to the calling broker.

Auth: Required. Caller must have tier = ENABLER.

Request body:

Field	Type	Required	Notes
ewrId	integer	'	Must be a TEA eWR; active mandate must exist from its owner to you
grade	string	'	e.g. "BOP", "BOPF", "PF", "D"
gradeMark	string	'	Estate mark, e.g. "NKT"
giOrigin	string	'	Geographic Indication, e.g. "Nyeri-ABK"
grossWeightKg	number	'	Total weight including packaging
netWeightKg	number	'	Tea weight only
tareWeightKg	number	'	Packaging/container weight
packageType	string	'	e.g. "sack", "chest"
packingWeightKg	number	—	Per-package weight
tasterRemarks	string	—	Cupping/tasting notes
certifications	string[]	—	e.g. ["ORGANIC", "FAIRTRADE", "RAINFOREST_ALLIANCE"]
storageStatus	string	—	e.g. "IN_STORE", "DISPATCHED"
listingType	"AUCTION" \	"FIXED_PRICE"	—
catalogueType	"WITH_VALUATION" \	"WITHOUT_VALUATION"	—
reservePriceUsd	number	' if AUCTION	Minimum sale price
brokerValuationUsd	number	—	Broker's indicative valuation
fixedPricePerKgUsd	number	' if FIXED_PRICE	Price per kg for spot listing
commissionRate	number	—	0–1 fraction, default 0.01 (1%). Overridden by mandate if set.
tickTiers	TickTier[]	—	See Tick Tiers. Default: [] (falls back to 1.5%)
antiSnipeConfig	AntiSnipeConfig	—	See Anti-Snipe. Default: {windowSecs:180, extensionSecs:180, maxExtensionSecs:1800}
bidSecurityPct	number	—	0–1 fraction, default 0.10 (10%)

Response: 201 Created — the created TeaLot record (status: DRAFT).

GET /tea/lots

List tea lots with optional filters.

Auth: Not required (public endpoint).

Query parameters:

Param	Type	Description
grade	string	Exact grade match
giOrigin	string	Exact GI origin match
certification	string	Lot must include this certification
listingType	AUCTION \	FIXED_PRICE
status	string	Filter by lot status

brokerId	integer	Filter by broker user ID
ownerId	integer	Filter by producer user ID

Response: 200 OK — array of TeaLot records.

GET /tea/lots/:lotId

Retrieve full lot detail, enriched with owner name, broker name, and linked eWR summary.

Auth: Not required.

Response: 200 OK

```
{
  "id": 42,
  "ewrId": 7,
  "ownerId": 3,
  "brokerId": 5,
  "ownerName": "Nyeri Tea Factory",
  "brokerName": "East Africa Brokers Ltd",
  "grade": "BOP",
  "gradeMark": "NKT",
  "giOrigin": "Nyeri-ABK",
  "grossWeightKg": "510.000",
  "netWeightKg": "500.000",
  "tareWeightKg": "10.000",
  "packageType": "sack",
  "certifications": ["RAINFOREST_ALLIANCE"],
  "listingType": "AUCTION",
  "reservePriceUsd": "2800.00",
  "commissionRate": "0.0100",
  "tickTiers": [{"upToUsd": 3000, "incrementPct": 1.5}, {"above": true, "incrementPct": 1.0}],
  "antiSnipeConfig": {"windowSecs": 180, "extensionSecs": 180, "maxExtensionSecs": 1800},
  "bidSecurityPct": "0.1000",
  "status": "CATALOGUED",
  "sessionId": null,
  "auctionStartAt": null,
  "auctionEndAt": null,
  "totalExtensionSecs": 0,
  "ewr": {
    "ewrsReceiptId": "KE-NBI-2024-007",
    "commodityType": "TEA",
    "weightMt": "0.500",
    "harvestSeason": "2024-Q1",
    "state": "MARKET_LISTED",
    "teaProcessingType": "CTC",
    "teaLeafGrade": "BOP",
    "teaInvoiceSerial": "INV-2024-007"
  }
}
```

PATCH /tea/lots/:lotId

Update a lot's metadata. Only allowed in DRAFT or CATALOGUED status. Callable by the mandate broker or the eWR owner.

Auth: Required. Caller must be the lot's broker or owner.

Request body: Any subset of the createTeaLotSchema fields (excluding ewrId). Same validation rules apply — if listingType is AUCTION, a reservePriceUsd must exist; if FIXED_PRICE, a fixedPricePerKgUsd must exist.

Side effect: If the lot is DRAFT and any field is updated, status automatically advances to CATALOGUED and publishedAt is set.

Response: 200 OK — updated TeaLot record.

Endpoints — Auction Sessions

POST /tea/auctions

Create a new auction session and assign lots to it in catalogue order.

Auth: Required. Caller must have tier = ENABLER or ADMIN.

Request body:

Field	Type	Required	Notes
auctionDate	string	'	YYYY-MM-DD format
lotIds	integer[]	'	Ordered list of lot IDs. All must be in CATALOGUED or DISPATCHED status. ENABLER callers must own all lots as mandate broker.

Response: 201 Created

```
{
  "id": 12,
  "createdByBrokerId": 5,
  "auctionDate": "2024-03-15",
  "catalogueOrder": [42, 43, 44],
  "currentLotId": null,
  "status": "SCHEDULED",
  "createdAt": "2024-03-14T10:00:00Z",
  "updatedAt": "2024-03-14T10:00:00Z"
}
```

GET /tea/auctions/:sessionId

Retrieve session detail including all lots in catalogue order, enriched with live bidding data.

Auth: Required. Accessible by: ADMIN, the creating broker, any lot owner/broker within the session, or any bidder who has placed a bid in the session.

Response: 200 OK

```
{
  "id": 12,
  "status": "LIVE",
  "auctionDate": "2024-03-15",
  "currentLotId": 43,
  "brokerName": "East Africa Brokers Ltd",
  "lots": [
    {
      "id": 42,
      "status": "SOLD",
      "currentHighBidUsd": 3100.00,
      "bidCount": 7,
      "secsRemaining": null,
      "minNextBidUsd": null
    },
    {
      "id": 43,
      "status": "LIVE",
      "currentHighBidUsd": 2950.00,
      "bidCount": 3,
      "secsRemaining": 312.4,
      "minNextBidUsd": 2994.25
    },
    {
      "id": 44,
      "status": "CATALOGUED",
      "currentHighBidUsd": null,
      "bidCount": 0,
      "secsRemaining": null,
      "minNextBidUsd": null
    }
  ]
}
```

POST /tea/auctions/:sessionId/start

Start a SCHEDULED session. Moves the first lot to LIVE status and sets its countdown timer. Broadcasts a tea_lot_live SSE event.

Auth: Required. Caller must be the creating broker or ADMIN.

Request body:

Field	Type	Required	Notes
durationMins	integer	—	Per-lot duration in minutes. Must be 1–120. Default: 7.

Response: 200 OK

```
{
  "sessionId": 12,
  "currentLotId": 42,
  "auctionEndAt": "2024-03-15T09:07:00Z"
}
```

Endpoints — Bidding & Settlement

POST /tea/lots/:lotId/bids

Place a bid on a LIVE lot. Enforces tick-tier minimum increments, anti-snipe extension, and bid security holds. Auth: Required. Caller must have tier = OFF_TAKER. The lot owner cannot bid on their own lot.

Request body:

Field	Type	Required	Notes
amountUsd	number	'	Must meet or exceed the tick-tier minimum (minNextBidUsd from GET session)

Behaviour:

- Previous winning bidder's hold is released; new bidder's hold is created for amountUsd × bidSecurityPct.
- If the bid arrives within antiSnipeConfig.windowSecs of auctionEndAt, the timer is extended by antiSnipeConfig.extensionSecs (capped by maxExtensionSecs).
- Runs inside a SERIALIZABLE transaction with SELECT ... FOR UPDATE on the lot row to prevent race conditions.

Response: 201 Created

```
{
  "bid": {
    "id": 99,
    "lotId": 43,
    "sessionId": 12,
    "bidderId": 8,
    "amountUsd": 2950.00,
    "isWinning": true,
    "placedAt": "2024-03-15T09:04:12Z"
  },
  "antiSnipeTriggered": false,
  "newEndAt": null,
  "bidSecurityHeldUsd": 295.00
}
```

Error responses:

Code	Condition
400	Lot is not LIVE, auction has ended, or bid is below the minimum
400	Response includes minBidUsd field when below-minimum
403	Caller is not OFF_TAKER, or is the lot owner

POST /tea/lots/:lotId/take-out

Broker withdraws a lot that closed with RESERVE_NOT_MET. Transitions the lot to WITHDRAWN and releases all remaining bid-security holds.

Auth: Required. Caller must be the mandate broker or ADMIN.

No request body required.

Response: 200 OK — updated TeaLot record with status: "WITHDRAWN".

POST /tea/lots/:lotId/accept-below-reserve

Broker accepts the current winning bid even though it did not meet the reserve price. Transitions the lot to SOLD, transfers eWR ownership, and creates a settlement record.

Auth: Required. Caller must be the mandate broker or ADMIN.

No request body required.

Response: 200 OK

```
{
  "lotId": 43,
  "status": "SOLD",
  "acceptedBelowReserve": true,
  "grossAmountUsd": 2700.00,
  "promptDate": "2024-03-29"
}
```

POST /tea/lots/:lotId/settle

Admin confirms that payment has been received for a SOLD lot. Sets paymentStatus to PAID, marks the delivery order as ISSUABLE, and releases the winning bidder's bid-security hold.

Auth: Required. Caller must have tier = ADMIN.

No request body required.

Response: 200 OK — updated TeaLotSettlement record.

GET /tea/lots/:lotId/settlement

Retrieve the settlement record for a sold lot, including financial breakdown and prompt date.

Auth: Required. Accessible by: ADMIN, the lot owner (producer), the lot's broker, or the winning buyer.

Response: 200 OK

```
{
  "id": 55,
  "lotId": 43,
  "sessionId": 12,
  "winningBidId": 99,
  "buyerId": 8,
  "buyerName": "Nairobi Tea Buyers Co.",
  "grossAmountUsd": "2950.00",
  "platformFeeUsd": "14.75",
  "brokerCommissionUsd": "29.50",
  "netProducerAmountUsd": "2905.75",
  "promptDate": "2024-03-29",
  "paymentStatus": "PENDING",
  "deliveryOrderStatus": "NOT_ISSUABLE",
  "acceptedBelowReserve": 0
}
```

Endpoints — Dispatch Documents

POST /tea/lots/:lotId/dispatch

Attach a dispatch document to a lot. Callable by the mandate broker or lot owner.

Auth: Required.

Request body:

Field	Type	Required	Notes
docType	string	'	One of PRE_AUCTION_DISPATCH, WEIGHMENT_REPORT, DELIVERY_ORDER

docData	object	—	Flexible JSONB payload (see below). Default: {}
---------	--------	---	---

docData payloads by type:

docType	Expected docData fields
PRE_AUCTION_DISPATCH	{ dispatchDate, warehouseCode, truckId, ... }
WEIGHTMENT_REPORT	{ netWeightKg, grossWeightKg, weighedBy, weighedAt, ... }
DELIVERY_ORDER	{ buyerId, issuedAt, dueDate, instructions, ... } — only allowed when lot status is SOLD

Side effect: Submitting a PRE_AUCTION_DISPATCH document on a CATALOGUED lot automatically advances it to DISPATCHED.

Response: 201 Created — the created TeaDispatchDoc record.

GET /tea/lots/:lotId/dispatch

List all dispatch documents attached to a lot, including the submitter's name.

Auth: Not required.

Response: 200 OK — array of documents:

```
[
  {
    "id": 11,
    "lotId": 43,
    "docType": "PRE_AUCTION_DISPATCH",
    "submittedBy": 5,
    "submitterName": "East Africa Brokers Ltd",
    "docData": { "dispatchDate": "2024-03-13", "warehouseCode": "NBI-WH-04" },
    "createdAt": "2024-03-13T14:22:00Z"
  }
]
```

Auction Mechanics

Tick Tiers

Tick tiers define the minimum bid increment as a percentage of the current high bid. They are configured per lot and evaluated from first to last.

Structure:

```
[
  { "upToUsd": 1000, "incrementPct": 2.0 },
  { "upToUsd": 3000, "incrementPct": 1.5 },
  { "above": true, "incrementPct": 1.0 }
]
```

Resolution: Walk the tiers in order. The first tier where currentHigh "d upToUsd (or above: true) applies. Minimum next bid = currentHigh × (1 + incrementPct / 100), rounded to 2 decimal places.

Fallback: If tickTiers is empty, a flat 1.5% increment is used.

Example with the tiers above and current high of \$2,950:

- \$2,950 "d \$3,000 !" 1.5% tier applies
- minNextBid = 2950 × 1.015 = \$2,994.25

Anti-Snipe

Prevents last-second bid sniping by extending the auction timer when a bid arrives near the end.

Configuration (per lot, antiSnipeConfig):

Field	Default	Description
-------	---------	-------------

windowSecs	180	If a bid arrives within this many seconds of auctionEndAt, the timer is extended
extensionSecs	180	Number of seconds added to auctionEndAt per triggered extension
maxExtensionSecs	1800	Total cap on all extensions (30 minutes). Once reached, no further extensions occur.

totalExtensionSecs on the lot tracks cumulative extension and is reset to 0 when a lot goes LIVE.

Settlement & Prompt Date

When a lot closes as SOLD, the system automatically:

1. Creates a TeaLotSettlement record with the financial breakdown.
2. Transfers eWR ownership to the buyer (ewrs.ownerId != buyer, ewrs.state != SETTLED).
3. Computes the Prompt Date — the payment deadline.

Fee breakdown (on gross bid amount):

Item	Rate	Example (\$2,950 gross)
Platform fee	0.5%	\$14.75
Broker commission	configured per lot (default 1%)	\$29.50
Net to producer	remainder	\$2,905.75

Prompt Date = trade date + 10 Kenyan working days, skipping weekends and the following public holidays:

Holiday	Date
New Year's Day	1 Jan
Labour Day	1 May
Madaraka Day	1 Jun
Huduma Day	10 Oct
Mashujaa Day	20 Oct
Jamhuri Day	12 Dec
Christmas Day	25 Dec
Boxing Day	26 Dec
Good Friday	Easter " 2 days
Easter Monday	Easter + 1 day

If payment is not confirmed by the Prompt Date, the background worker marks the settlement DEFAULTED, forfeits the buyer's bid-security hold, and deducts 15 reputation points from the buyer's account.

Real-Time Events (SSE)

The Tea Marketplace subscribes to GET /api/auctions/stream?sessionId=:id (shared with the WRS auction stream). Events are delivered as Server-Sent Events with the following shapes:

Event type	Trigger	Payload
tea_lot_live	Session started or next lot promoted	{ sessionId, lotId, status:"LIVE", commodity:"TEA", auctionEndAt }
bid	Bid placed	{ lotId, sessionId, commodity:"TEA", bid:{id, amountUsd, bidderId, placedAt}, antiSnipeTriggered, newEndAt, bidSecurityHeldUsd }

closed	Lot sold	{ lotId, status:"SOLD", commodity:"TEA", sessionId, winningBidId, grossAmountUsd, promptDate }
reserve_not_met	Lot closed below reserve	{ lotId, status:"RESERVE_NOT_MET", commodity:"TEA", sessionId }
session_closed	All lots in session complete	{ sessionId, status:"CLOSED", commodity:"TEA" }

Events are also published to PostgreSQL pub/sub (pg_notify) on the auction_events channel, enabling multi-instance fan-out.

Note: Bidder identity in the Tea Marketplace UI is anonymised (displayed as "Buyer A", "Buyer B", etc.) even though bidderId is present in the raw SSE payload.

Background Worker

File: artifacts/api-server/src/lib/tea-auction-worker.ts

Runs on a 10-second polling interval. Uses pg_try_advisory_lock(3) so only one server instance drives transitions at a time (the others skip their tick if the lock is held).

Phase 1 — Close expired lots:

Finds all lots with status = 'LIVE' and auctionEndAt "d now. For each:

- Reserve met !' SOLD: creates settlement record, transfers eWR ownership, releases non-winner holds.
- Reserve not met !' RESERVE_NOT_MET: releases non-winner holds, keeps winner's hold in case broker accepts below reserve.
- After closing, advances the session to the next catalogue lot (sets it LIVE for 7 minutes), or marks the session CLOSED if all lots are done.

Phase 2 — Forfeit defaulted bid security:

Finds settlements with paymentStatus = 'PENDING' and promptDate < today. For each:

- Sets hold status to FORFEITED.
- Sets settlement paymentStatus to DEFAULTTED.
- Deducts 15 reputation points from the buyer.

Database Schema

tea_lots

Column	Type	Notes
id	serial PK	
ewr_id	integer FK !' ewrs.id	Must be a TEA eWR
owner_id	integer FK !' users.id	Tea producer
broker_id	integer FK !' users.id	Mandate broker
grade	text	e.g. "BOP", "BOPF", "PF"
grade_mark	text	Estate mark, e.g. "NKT"
gi_origin	text	Geographic Indication
gross_weight_kg	numeric(12,3)	
net_weight_kg	numeric(12,3)	
tare_weight_kg	numeric(12,3)	
package_type	text	

packing_weight_kg	numeric(10,3)	Per-package weight, optional
taster_remarks	text	Optional cupping notes
certifications	jsonb	String array, e.g. ["ORGANIC"]
storage_status	text	Optional
listing_type	enum	AUCTION \
catalogue_type	enum	WITH_VALUATION \
reserve_price_usd	numeric(14,2)	Required for AUCTION
broker_valuation_usd	numeric(14,2)	Optional
fixed_price_per_kg_usd	numeric(10,4)	Required for FIXED_PRICE
commission_rate	numeric(5,4)	Default 0.0100 (1%)
tick_tiers	jsonb	Array of TickTier objects
anti_snipe_config	jsonb	{windowSecs, extensionSecs, maxExtensionSecs}
bid_security_pct	numeric(5,4)	Default 0.1000 (10%)
session_id	integer	Set when assigned to a session
auction_start_at	timestampz	When this lot went LIVE
auction_end_at	timestampz	Current deadline (extended by anti-snipe)
total_extension_secs	integer	Cumulative anti-snipe extension
status	enum	DRAFT \
published_at	timestampz	Set when lot moves from DRAFT !' CATALOGUED
created_at / updated_at	timestampz	

tea_auction_sessions

Column	Type	Notes
id	serial PK	
created_by_broker_id	integer FK !' users.id	
auction_date	date	Planned date YYYY-MM-DD
catalogue_order	jsonb	Ordered integer array of lot IDs
current_lot_id	integer	Lot currently being auctioned (no FK to avoid circular dep)
status	enum	SCHEDULED \
created_at / updated_at	timestampz	

tea_lot_bids

Column	Type	Notes
id	serial PK	
lot_id	integer FK !' tea_lots.id	
session_id	integer FK !' tea_auction_sessions.id	
bidder_id	integer FK !' users.id	
amount_usd	numeric(14,2)	

is_winning	boolean	true for the current high bid only
placed_at	timestampz	

tea_lot_settlements

Column	Type	Notes
id	serial PK	
lot_id	integer FK !' tea_lots.id	Unique — one settlement per lot
session_id	integer FK !' tea_auction_sessions.id	
winning_bid_id	integer FK !' tea_lot_bids.id	
buyer_id	integer FK !' users.id	
gross_amount_usd	numeric(14,2)	
platform_fee_usd	numeric(14,2)	0.5% of gross
broker_commission_usd	numeric(14,2)	commissionRate × gross
net_producer_amount_usd	numeric(14,2)	gross " platform fee " commission
prompt_date	date	Payment deadline (trade date + 10 Kenyan working days)
payment_status	enum	PENDING \
delivery_order_status	enum	NOT_ISSUABLE \
accepted_below_reserve	integer	1 if broker accepted below reserve; 0 otherwise
created_at / updated_at	timestampz	

tea_dispatch_docs

Column	Type	Notes
id	serial PK	
lot_id	integer FK !' tea_lots.id	
doc_type	enum	PRE_AUCTION_DISPATCH \
submitted_by	integer FK !' users.id	
doc_data	jsonb	Flexible payload; shape depends on doc_type
created_at	timestampz	

Fee Structure

Fee	Rate	Recipient
Platform fee	0.5% of gross bid	TokenHarvest platform
Broker commission	1% default (overridable by mandate or lot)	Mandate broker

Bid security hold	10% default (configurable per lot)	Held from buyer; released on payment or forfeited on default
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Error Reference

All errors return JSON with an "error" string. Validation failures additionally include an "issues" array (Zod format).

HTTP Code	Common causes
400	Invalid input, wrong lot status for the operation, bid below minimum, durationMins out of range
401	No Clerk session cookie / token
403	Wrong role (e.g. OFF_TAKER trying to create a lot), no active mandate, bidding on own lot
404	Lot, session, eWR, or user not found
500	Unexpected server error (details logged server-side)

When a bid is below the minimum, the 400 response includes an additional "minBidUsd" field:

```
{
  "error": "Bid must be at least $2994.25 (minimum increment above current high of $2950.00)",
  "minBidUsd": 2994.25
}
```


